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Hurdle Rate Buffers and Bargaining Power in Asset Acquisition

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1 Big picture

- **Question.** Why do firms use hurdle rates that are well above their cost of capital, and do these buffers necessarily destroy value by causing firms to pass up positive-NPV projects?
- **Setting.** The paper combines six CFO surveys, Compustat-based realized return measures, M&A fairness-opinion discount rates, and customer-supplier data to study the consequences of hurdle-rate buffers.
- **Core challenge.** Textbook capital budgeting treats project inputs as exogenous and therefore views a hurdle-rate buffer mainly as a filter that rejects some good projects. But in real project development, many inputs are bargained over, so a tougher hurdle can act as a commitment device in negotiation.
- **Main theoretical result.** An elevated hurdle rate can improve a firm's bargaining position during project development or acquisition negotiations. The gain from better terms can outweigh the loss from rejecting moderately positive-NPV projects.
- **Main empirical result.** In M&A data, higher bidder hurdle-rate buffers are associated with lower premia paid to targets and higher bidder announcement returns, but not with larger combined-firm gains. This is consistent with surplus capture rather than selection of fundamentally better deals.
- **Additional evidence.** In CFO survey and Compustat data, hurdle rates are persistent, set from the top of the organization, and realized ROIC bunches just above the hurdle rate rather than above WACC. This supports the idea that hurdle rates are a real internal benchmark, not just survey noise.
- **Key magnitudes.**
 - 78% of firms use a positive hurdle-rate buffer.
 - Conditional on a positive buffer, the mean hurdle rate is 14.73% and the mean WACC is 8.15%, for a mean buffer of 6.58 percentage points.
 - In the full survey sample, the average hurdle rate is 13.88%, average WACC is 8.77%, and the average intensive-margin buffer is 5.11 percentage points.
 - In the M&A sample, the implied buffer from fairness opinions is about 4.7 percentage points on average.
- **Economic takeaway.** Hurdle-rate buffers do not merely ration investment. They can also be part of an organizational bargaining technology: by committing project managers to a tough benchmark, the firm can negotiate better terms with suppliers, customers, or targets.

2 Data and measurement (key objects)

2.1 CFO survey data

- **Primary source.** The core survey evidence comes from six Duke CFO surveys: 2011q1, 2012q2, 2017q2, 2017q3, 2019q1, and 2022q2.
- **Key survey questions.** CFOs are asked directly:

- “What do you estimate is your firm’s overall weighted average cost of capital (WACC)?”
- “What is your firm’s hurdle rate (the rate of return that an investment must beat in order to be adopted)?”
- **Main variable.** The hurdle-rate buffer is

$$\text{Buffer}_i = \text{Hurdle Rate}_i - \text{WACC}_i.$$

- **Extensive vs. intensive margin.**
 - Extensive margin: indicator for whether the firm uses a positive buffer.
 - Intensive margin: the size of the buffer in percentage points.

2.2 Other datasets used in the paper

- **Compustat quarterly.** Used to construct realized return on invested capital (ROIC) for public firms that appear in the CFO surveys.
- **Refinitiv SDC Platinum M&A data.** Used for the acquisition analysis. The paper focuses on public-to-public transactions with fairness opinions reporting maximum and minimum discount rates.
- **CRSP and Compustat.** Used to estimate bidder and target abnormal returns and to construct firm controls such as size, leverage, cash, and market-to-book.
- **FactSet Revere.** Used to construct supplier networks and supplier-level bargaining-power proxies.
- **Compustat Segments / SEC customer disclosures.** Used to construct customer concentration.
- **Kim and Kung (2017).** Used for asset redeployability, an industry-level proxy for asset specificity.

2.3 Key descriptive objects

- **Sample size.** The CFO survey sample contains 1,232 observations.
- **Average levels (full survey sample).**
 - Hurdle rate: 13.879%.
 - Cost of capital (WACC): 8.769%.
 - Buffer, intensive margin: 5.111 percentage points.
 - Buffer, extensive margin: 0.777 (so 77.7% use a positive buffer).
- **Conditional on a positive buffer.**
 - Hurdle rate: 14.732%.
 - WACC: 8.153%.
 - Buffer: 6.579 percentage points.

- **Industry-level proxy means.**
 - Beta volatility: 0.686.
 - Sales volatility: 0.120.
 - Supplier concentration: 0.246.
 - Customer concentration: 0.305.
 - Supplier relative markup: -0.066.
 - Asset redeployability: 0.440.

2.4 Unobservables and timing

- **Why buffers exist.** The paper emphasizes that buffers may arise for many reasons already discussed in the literature: financing constraints, managerial/resource constraints, project prioritization, idiosyncratic risk or uncertainty, over-optimism, agency frictions, or real-options logic.
- **Who sets the hurdle.** The hurdle rate is typically chosen by upper-level management, not by the line manager bargaining over project inputs. A 2024q4 CFO survey reports that only about 3.6% of firms say middle management had a say in setting the hurdle.
- **Interpretation.** This top-down assignment is what gives the hurdle rate commitment value. Delegates in the organization take the hurdle as given and negotiate around it.
- **Timing in the bargaining story.**
 - The CFO sets the hurdle rate.
 - Project managers or M&A teams bargain with counterparties over project inputs or acquisition terms.
 - Realized returns then reflect both the project and the terms negotiated.

3 Models and empirical specifications (all equations + interpretation)

3.1 Project evaluation versus project development

- The paper's conceptual distinction is between:
 - **project evaluation:** the textbook case where inputs are exogenous and the hurdle just screens projects,
 - **project development:** a richer environment where project inputs are bargained over and the hurdle can affect negotiated terms.
- Example: if a plant requires land, the price of land is not exogenous. It is negotiated, and that negotiated price depends on the firm's willingness to walk away.

3.2 Delegated bargaining environment

The firm F has a true cost of capital

$$W_F = 1 + r_F^{wacc},$$

and the CFO may report either the true hurdle $H_F = W_F$ or a buffered hurdle $H_F = \tau_F > W_F$.

The outside counterparty O has true cost of capital

$$W_O = 1 + r_O^{wacc},$$

and reported hurdle $H_O \in \{W_O, \tau_O\}$.

Projects have gross return R drawn from a distribution $F(R)$ on $[1, \bar{R}]$.

If the project is feasible, the split of surplus going to firm F is

$$\theta(H_F, H_O),$$

with

$$\frac{\partial \theta}{\partial H_F} > 0, \quad \frac{\partial^2 \theta}{\partial H_F^2} \leq 0, \quad \frac{\partial \theta}{\partial H_O} < 0.$$

A convenient example used in the paper is the Tullock contest form:

$$\theta = \frac{H_F}{H_F + H_O}.$$

Interpretation. A higher reported hurdle rate improves the firm's bargaining position by making its delegate appear tougher or more willing to walk away.

3.3 Feasibility and the basic trade-off

A project is feasible only if it can satisfy both parties' required returns, so the minimum feasible return is

$$R = H_F + H_O.$$

If the CFO inflates the hurdle rate, two things happen:

- the firm's surplus share goes up,
- but some moderately good projects are no longer feasible.

This is the paper's main trade-off.

3.4 Single-firm optimal buffer

If the outside party does *not* strategically use a buffer, then without a buffer the firm's value is

$$V_N = \int_{\bar{R}'}^{\bar{R}} \theta_N dF(R) = \theta_N [1 - F(\bar{R}')],$$

where

$$\bar{R}' = W_F + W_O.$$

With a buffer, the firm solves

$$\max_{\tau_F \geq W_F} V_B = \int_{\bar{R}''}^{\bar{R}} \theta_B dF(R),$$

where

$$\bar{R}'' = \tau_F + W_O, \quad \theta_B = \theta(\tau_F, W_O).$$

Proposition 1. There exists a unique optimal buffered hurdle rate τ^* satisfying

$$\frac{\partial \theta_B(\tau^*) / \partial \tau}{\theta_B(\tau^*)} = \frac{f(\bar{R}'')}{1 - F(\bar{R}')}.$$

Interpretation.

- Left-hand side: marginal bargaining gain from raising the hurdle.
- Right-hand side: hazard-rate style marginal loss from cutting off one more project.

The firm uses a higher buffer when bargaining gains are large, starting bargaining power is weak, and the loss from excluding projects is relatively small.

3.5 Closed-form example under uniform returns

Suppose returns are uniform on $[1, \bar{R}]$ and the split is the Tullock contest function. Then:

$$\tau_F^* = \sqrt{\bar{R}W_O} - W_O,$$

$$V_B^* = \bar{R} + W_O - 2\sqrt{\bar{R}W_O},$$

and the gain from using the buffer is

$$\Delta V = V_B^* - V_N = \frac{W_O \bar{R}}{W_F + W_O} + (W_F + W_O) - 2\sqrt{\bar{R}W_O} > 0.$$

A positive buffer is optimal if

$$(1 - \theta_N) \bar{R} > W_F + W_O.$$

Interpretation. The firm is more likely to use a positive buffer when its ex ante bargaining power is weak (low θ_N), its own WACC is low, and there is more upside in the project distribution.

3.6 Competitive buffers and the prisoner's-dilemma logic

If both parties choose buffers strategically, the problem becomes

$$\max_{\tau_F} V_F = \int_{\bar{R}''}^{\bar{R}} \theta dF(R), \quad \max_{\tau_O} V_O = \int_{\bar{R}''}^{\bar{R}} (1 - \theta) dF(R),$$

with

$$\bar{R}'' = \tau_F + \tau_O.$$

Proposition 2. In any Nash equilibrium,

$$\frac{\partial \tau_F^*}{\partial \tau_O^*} = 1 - \frac{1}{\theta(\tau_F^*, \tau_O^*)}.$$

Under uniform $F(R)$ and Tullock splitting:

$$\tau_F^* = \tau_O^* = \frac{\bar{R}}{4}, \quad V_F^* = V_O^* = \frac{\bar{R}}{4}.$$

Deadweight loss relative to no buffers is

$$Loss = \frac{\bar{R}}{2} - (W_F + W_O).$$

Interpretation. Once one side uses a buffer, the other side also wants to harden its negotiating stance. This escalates hurdle rates, destroys some feasible projects, and can generate a bargaining arms race.

3.7 Many counterparties

If there are n counterparties and each chooses strategically, the paper shows:

$$\tau^* = \frac{(n-1)\bar{R}}{n^2}, \quad V^* = \frac{\bar{R}}{n^2}.$$

Interpretation. The equilibrium hurdle-rate buffer is strictly decreasing in the number of parties at the table. When many counterparties are involved, each individual agent’s leverage from exaggerating its walkaway threshold becomes smaller.

3.8 Model calibration and value implications

Figure 5 calibrates the model using CFO-survey moments:

- firm’s WACC: 8.77%,
- outside option WACC: 13.88% (mean and 90th percentile inputs in the calibration),
- targeted optimal buffer: 5.11 percentage points.

The paper compares:

- a standard capital-budgeting view that ignores bargaining,
- a bargaining view that includes better negotiated terms.

At the calibrated optimum:

- traditional capital-budgeting logic implies about -4.4% value loss,
- the full bargaining model implies about $+0.11\%$ value gain,
- the difference is about 4.5 percentage points.

Interpretation. This does not mean buffers are always good; it means that once bargaining is recognized, the blanket statement “buffers destroy value” is no longer correct.

3.9 ROIC bunching specification

The paper defines

$$ROIC_{i,t} = \frac{EBIT_{i,t}(1 - tax_{i,t})}{ICAPT_{i,t-1}},$$

where invested capital is the sum of long-term book debt and equity, and then studies

$$Excess\ ROIC_{i,t} = ROIC_{i,t} - Hurdle_{i,t}.$$

The idea is simple: if firms actually organize projects around the hurdle rate, realized ROIC should bunch just above the hurdle, not merely around WACC.

3.10 M&A empirical specification

For each deal, the paper defines the bidder's implied hurdle rate as the average of the maximum and minimum fairness-opinion discount rates. The implied buffer is

$$Implied\ Buffer = Average\ Fairness-Opinion\ Discount\ Rate - Predicted\ Target\ WACC.$$

The paper then estimates regressions of the form

$$Outcome_d = \beta Implied\ Buffer_d + \gamma Predicted\ Target\ WACC_d + Controls_d + \varepsilon_d,$$

where the outcomes are:

- premium paid to the target,
- bidder cumulative abnormal return (CAR),
- target CAR,
- combined-firm CAR,
- bidder share of dollar abnormal returns.

Bidder CAR is measured using the 4-factor model in the baseline over the event window $[-1, 1]$.

3.11 Bargaining-power empirical specification

Using survey firms matched to bargaining-power proxies, the paper estimates reduced forms such as

$$Buffer_{i,t} = \beta BargainingPower_{i,t} + \gamma WACC_{i,t} + X'_{i,t}\delta + \varepsilon_{i,t},$$

for both:

- the extensive margin of buffer use,
- the intensive margin of buffer size.

The bargaining-power variables are:

- supplier concentration,

- customer concentration,
- supplier relative markup.

Controls include cost of capital, beta volatility, sales volatility, and fixed effects for survey quarter, size, ownership, and credit rating in richer specifications.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard interpretations miss

- Standard explanations of buffers focus on why firms might want a tougher hurdle – financial constraints, risk, agency, or managerial bottlenecks.
- But those explanations do not directly explain why large and persistent buffers would survive if they only caused firms to leave value on the table.
- The paper’s contribution is to identify a *consequence* of high hurdles: they affect bargaining outcomes.

4.2 What identifies the bargaining channel in M&A

- The key object is the fairness-opinion discount rate, which the paper treats as a proxy for the bidder’s hurdle rate in the deal, following Dessaint et al. (2021).
- If higher buffers simply reflected better target selection, then higher buffers should predict higher *combined* merger gains.
- Instead, the paper finds that higher buffers predict:
 - lower deal premia,
 - higher bidder CAR,
 - no meaningful increase in combined CAR.
- This pattern is consistent with a redistribution of surplus toward the bidder, not with higher underlying synergies.

4.3 What identifies the ex ante bargaining-power results

- Supplier concentration, customer concentration, and supplier relative markup are all proxies for the firm’s bargaining position relative to trading partners.
- The sign prediction is clear: firms with less ex ante bargaining power should have more reason to use a buffer.
- The regressions control for cost of capital and for volatility-based variables that proxy for some standard reasons for buffers. This helps separate the bargaining channel from a pure risk or uncertainty channel.

- The paper further sharpens the interpretation using asset redeployability: the bargaining-buffer relation should be stronger when assets are more specific and bargaining matters more.

4.4 Why ROIC bunching matters

- The ROIC bunching evidence does not by itself prove the bargaining model.
- But it is powerful supporting evidence that hurdle rates are a real internal benchmark that shapes action. Without this fact, it would be harder to interpret the fairness-opinion and bargaining-power results as consequences of organizational commitment.
- The additional finding that there is no analogous bunching around WACC makes the result especially compelling.

4.5 Relation to the literature

- **Capital budgeting / hurdle-rate literature:** The paper complements work explaining why buffers exist by showing how they can affect project development and value.
- **Bargaining and delegation:** The model follows the idea that delegation and commitment can change negotiated outcomes.
- **M&A and discount-rate practice:** The paper connects internal capital-budgeting rules to observable merger outcomes.

5 Tables and figures

5.1 Table 1 and Figure 1: prevalence and reasons for buffers

Table 1

Summary statistics.

This table displays the number of observations, averages, standard deviations, and quartiles of key empirical variables. Panel A displays statistics for variables related to the hurdle rate and the cost of capital. Panel B displays summary statistics for the industry-level variables we explore in Section 5. Summary statistics for markup variables are displayed only for firms in non-consumer-facing industries. Detailed variable definitions are available in [Table B.1](#).

	N	Mean	Std Dev	25%	50%	75%
Panel A: CFO survey variables						
Hurdle rate	1232	13.879	5.977	10	12.750	16
Cost of capital (WACC)	1232	8.769	4.211	5.500	8.800	11
Buffer (Extensive margin)	1232	0.777				
Buffer (Intensive margin)	1232	5.111	5.376	0.787	4	7.500
Hurdle rate Buffer > 0	957	14.732	6.072	10	15	18
WACC Buffer > 0	957	8.153	3.905	5	8	10
Buffer Buffer > 0	957	6.579	5.248	3	5	9.500
Panel B: Industry-level variables						
Beta volatility	1232	0.686	0.234	0.535	0.687	0.834
Sales volatility	1232	0.120	0.053	0.088	0.120	0.154
Supplier concentration	1066	0.246	0.066	0.204	0.249	0.282
Customer concentration	1066	0.305	0.191	0.136	0.333	0.428
Supplier relative markup	1066	-0.066	0.080	-0.089	-0.071	-0.022
Asset redeployability	1232	0.440	0.128	0.346	0.476	0.516

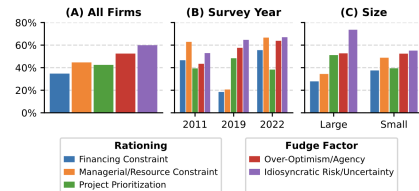


Fig. 1. Reasons that companies set hurdle rates above the cost of capital. This figure displays qualitative reasons why firms set their hurdle rates above WACC (i.e., use a hurdle rate buffer). Results are taken from three editions of the Duke CFO survey (2011q1, 2019q1, 2022q2) which asked CFOs “Why do you set your hurdle rate above WACC?” Possible answers vary by survey, so we group them into five distinct reasons. Table 5.2 displays these groupings and provides further detail on how the question was asked on each survey. “Rationing” refers to cases when managers prioritize projects, for example due to limited capital resources. “Fudge Factor” refers to cases when managers add a (positive) buffer to their discount rate to account for, e.g., idiosyncratic risk of specific projects (Brealey and Myers, 1996; Dechow, 2024). Panel A displays the percentage of CFOs that fall within each category. Panels B and C display the results split by survey year and size (revenue above/below \$1 billion).

Read:

- Table 1 is the first anchor: the typical firm uses a large hurdle-rate buffer. The overall mean hurdle is 13.88%, the mean WACC is 8.77%, and 77.7% of firms use a positive buffer.

- Among firms with a positive buffer, the average buffer is 6.58 percentage points. This is economically huge in present-value terms.
- Figure 1 shows that firms themselves report many reasons for using buffers. About 42% mention project prioritization, and about 60% mention risk or uncertainty in the discount rate or project environment.
- The paper’s message is not that bargaining replaces these explanations. Rather, even if the buffer exists for one of these “non-strategic” reasons, it can still improve bargaining outcomes.

5.2 Figure 2 and Figure 3: variation and persistence of hurdle rates

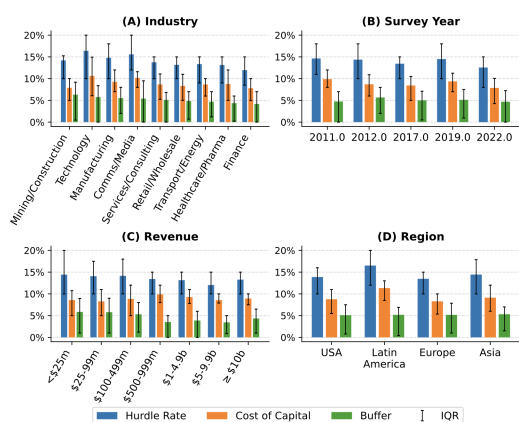


Fig. 2. Time series and cross-sectional variation in hurdle rates and the cost of capital. This figure displays the within-group average and interquartile range (IQR) of CFO survey company hurdle rates, WACCs and hurdle rate buffers across several observable characteristics. For example, Panel A displays the statistics across CFO survey industries; the average buffer for firms in Mining/Construction is about 6%, and the 25th and 75th percentiles within-group are roughly 0% and 9%, respectively.

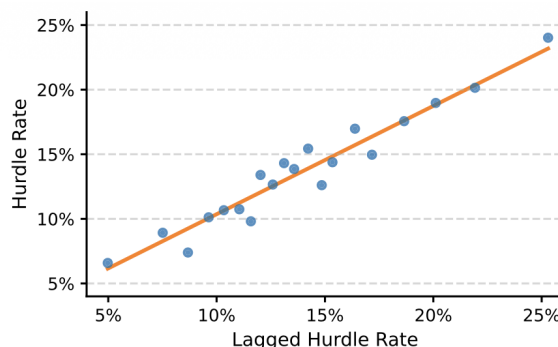


Fig. 3. Sticky hurdle rates.

This figure displays persistence of hurdle rates within-firm in our CFO survey data by displaying a binned scatterplot of the current hurdle rate regressed on the lagged hurdle rate. To be included in this analysis, a CFO must appear in at least two of the six surveys we use in our analysis. The slope coefficient is about 0.84 in a regression that includes beta volatility, sales volatility controls, along with industry, year and size category fixed effects.

Read:

- Figure 2 shows that buffers vary across industries, survey years, size groups, and regions, but are positive in all of them.
- A particularly useful fact is that smaller firms have larger buffers: about 5.8% for firms with revenue below \$1 billion versus about 3.8% for larger firms.
- Figure 2 also shows that the buffer is relatively stable through time, even though hurdle rates and WACC move across groups.
- Figure 3 sharpens this: the within-firm persistence slope is about 0.84 after controls. Nearly half of firms in consecutive surveys report no hurdle-rate change.
- This persistence is important for the theory because a commitment device must actually be stable enough that delegates believe it.

5.3 Figure 4: ROIC bunching just above the hurdle

Read:

- Figure 4 is the cleanest evidence that the hurdle is a focal internal benchmark.

- The distribution of excess ROIC, defined as ROIC minus the hurdle rate, shows a clear discontinuity at zero.
- About 9% of relevant observations lie in the +0.5% bin, compared with only about 3.5% in the −0.5% bin.
- The paper finds no comparable bunching around WACC.
- Interpretation: firms appear to organize projects to “meet or beat” the hurdle itself, not merely the cost of capital.

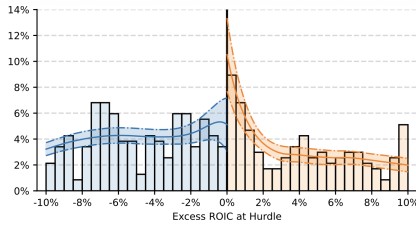


Fig. 4. Density manipulation in excess ROIC at the hurdle rate. This figure displays the results of testing for density manipulation of return on invested capital (ROIC) around the hurdle rate, following the methodology described in Cattaneo et al. (2018). The sample is CFO survey firms that also appear in Compustat quarterly, for which we observe a hurdle rate. We first linearly interpolate firms’ hurdle rates between two consecutive appearances in Compustat. For example, if a firm’s hurdle rate in 2012q2 were 10%, and in 2012q4 it were 11%, we would assume that the value of the hurdle rate in 2012q3 were 10.5%. Given stickiness in hurdle rates (as in, e.g., Graham, 2002), this assumption seems reasonable. ROIC is defined as $ROIC_{it} = EBIT_{it}(1 - \tau_{tax})/ICAP_{it}$, where $ICAP_{it}$ is the firm’s invested capital (i.e., the sum of long-term book debt and equity). We take trailing four-quarter sums of $EBIT(1 - \tau_{tax})$ and divide by $ICAP_{it}$ from the previous fiscal year. Next, we define our variable of interest, “Excess ROIC”, as a firm’s ROIC above its hurdle rate, i.e. for firm i in quarter t , $Excess\ ROIC_{it} = ROIC_{it} - Hurdle_{it}$. We focus on observations of Excess ROIC with an absolute value less than 25%. The sample is 470 observations. We are interested in testing for density manipulation of Excess ROIC at 0. The blue and orange bars display observed frequencies of Excess ROIC in 0.5% bins ranging from −10% to +10% (though the densities estimated use the full range of observations). The blue and orange lines (and shaded areas) display the estimated local-polynomial densities as described in Cattaneo et al. (2018). To estimate the densities, we set the order of the local polynomial (and the order of the bias-corrected density estimator) equal to 3. The bandwidths for the local polynomial estimator are chosen via the “data-driven” methodology described in Section 2.5 of Cattaneo et al. (2018), and are $(h, h_1) = (7.449\%, 1.19\%)$. The shaded areas display 90% confidence intervals. Table IA.2 displays robustness where we vary the size of the bandwidth and the order of the polynomials. The figure displays a discontinuity of the ROIC density at the hurdle rate, with some mass of the ROIC density shifting from just below to just above the hurdle rate. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

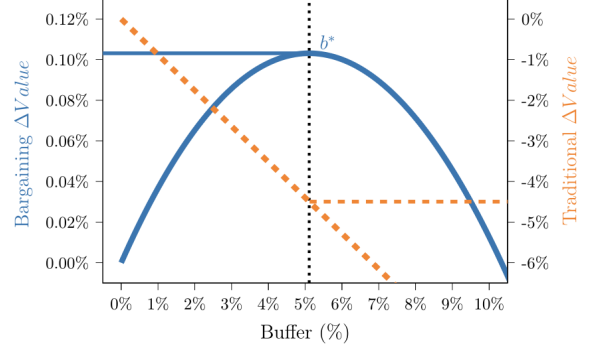


Fig. 5. Implications for firm value from the delegated bargaining model.

This figure plots the model-implied change in firm value from using a hurdle rate buffer and the change in firm value from using a buffer as implied by a standard capital budgeting model. Using Corollary 1, we calibrate our model such that $r_F^{WACC} = 8.77\%$ and $r_O^{WACC} = 13.88\%$, the mean and 90th percentile WACC in the CFO Survey data. We choose remaining parameters such that the optimal hurdle rate buffer $b^* = 5.11\%$, which is the average buffer in our sample (given r_F^{WACC} and r_O^{WACC} , $R = 2.226$ and we set $\bar{R} = 4.555$, about $2 \times R$). Under these parameters, the percentage increase in firm value from the no-buffer case to the optimal buffered hurdle rate, $V_B^*/V_N - 1 \approx 0.11\%$, and is displayed in blue relative to the left y-axis. Displayed on the right y-axis is the percentage change in value if we were to ignore the benefits from bargaining (and only consider the value lost from forgoing positive NPV projects), which is roughly -4.4% . Thus, the difference in value implications once we consider the effects of bargaining is about 4.5 percentage points. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.4 Figure 5: why a buffer need not destroy value

Read:

- Figure 5 translates the theory into the paper’s core economic point.
- If we ignore bargaining and only count forgone positive-NPV projects, the calibrated buffer of 5.11 percentage points looks quite harmful: about a 4.4% value loss.
- Once bargaining gains are included, the same buffer increases firm value by about 0.11%.
- The difference – roughly 4.5 percentage points – is the paper’s headline correction to the textbook intuition.

5.5 Table 2 and Table 3: merger outcomes and ex post surplus capture

Read:

Table 2

Hurdle rate buffers and negotiation outcomes: Evidence from M&A deals.

This table presents analysis of the relation between a bidder's use of a buffer and bidder bargaining outcomes in M&A deals. The sample is public-to-public M&A deals with maximum and minimum discount rates used in fairness opinions on proposed deals. Following Desai et al. (2021), we compute the average of the maximum and minimum discount rate used in the fairness opinion. The variable Implied Buffer is the difference between the fairness opinion average discount rate and the target's predicted WACC (see Desai et al., 2021). In Panel A, Premium is the percentage premium of the bid-implied equity value over the target's market capitalization in the year prior to the bid. Bidder CAR is the bidder's cumulative abnormal return relative to the 4-factor model over the 3 days around deal announcement $t \in [-1, 1]$. In Panel B, Target CAR is the same for the target. Combined CAR is the combined-firm CAR, weighted by the bidder and target market capitalization in the year prior to the bid. Bidder Share SCAR refers to the bidder's share of the total dollar abnormal return (Desai, 2012), which is only well-defined for deals with weakly positive abnormal returns for both firms. Year FE refers to the year the deal was announced. Bidder/Target Industry FE refers to the bidder/target France-French 49 industry code. Bidder/Target controls are lag market capitalization in the year prior to bid, market-to-book, return on assets, leverage, and cash/assets. Deal controls are relative size (deal value to bidder market cap), toehold (percent owned by bidder before bid), indicators for deal type (cash, stock, mixed), hostile, same industry, tender offer and poison (see Desai et al. (2021) for variable definitions of controls). Detailed variable definitions are in Table B.1. Standard errors are clustered by 3-digit target SIC industry and displayed below the coefficient. ***, **, * denote significance at 1%, 5%, 10%.

Panel A: Elevated hurdle rates, Deal premia and Bidder CAR	(1)	(2)	(3)	(4)
	Premium	Bidder CAR		
Implied buffer	-1.456*** (0.446)	-1.218*** (0.405)	0.220** (0.085)	0.221*** (0.081)
Predicted target WACC	-1.155 (0.918)	-0.900 (0.823)	-0.100 (0.235)	-0.082 (0.245)
Observations	736	736	736	736
R-squared	0.406	0.460	0.225	0.242
Year FE	Yes	Yes	Yes	Yes
Bidder & target industry FE	Yes	Yes	Yes	Yes
Bidder & target controls	Yes	Yes	Yes	Yes
Target takeover defense controls	Yes	Yes	Yes	Yes
Deal controls	Yes	Yes	Yes	Yes

Panel B: Elevated hurdle rates and combined synergy gains	(1)	(2)	(3)	(4)	(5)	(6)
	Target CAR	Combined CAR		Bidder share SCAR		
Implied buffer	-0.640 (0.446)	-0.633 (0.415)	0.121 (0.132)	0.164 (0.147)	1.374** (0.650)	1.254* (0.654)
Predicted target WACC	-0.452 (0.440)	-0.329 (0.452)	-0.329 (0.339)	-0.332 (0.332)	0.681 (1.762)	0.655 (1.769)
Observations	736	736	736	736	284	284
R-squared	0.363	0.390	0.276	0.296	0.518	0.528
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Bidder & target industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Bidder & target controls	Yes	Yes	Yes	Yes	Yes	Yes
Target takeover defense controls	Yes	Yes	Yes	Yes	Yes	Yes
Deal controls	Yes	Yes	Yes	Yes	Yes	Yes

Table 3

Bidder abnormal returns and the relative costs of capital of bidder and target.

This table displays analysis in which we test the relation between the bidder's hurdle rate buffer and the bidder's CAR, conditional on the relative size of the bidder and target's WACC in the year prior to deal announcement. The variable "Implied Buffer" is the difference between the bidder's implied hurdle rate from the fairness opinion and the target WACC. The variable 1 [Bidder WACC > Target WACC] takes a value of one if the bidder's WACC is larger than the target's WACC. In line with the model's prediction, the bidder has less ex ante bargaining power when this takes a value of zero, so the model predicts a positive relation between the buffer and bidder CAR. The variables Target WACC and Bidder WACC are standardized to zero mean, unit variance. The controls are the same as used in Table 3. Standard errors are clustered by 3-digit target SIC industry and displayed below the coefficient. ***, **, * denote significance at 1%, 5%, 10%.

Table 3	(1)	(2)	(3)	(4)	(5)
	Bidder CAR				
Implied buffer	0.271*** (0.079)	0.344*** (0.090)	0.390*** (0.092)	0.327*** (0.096)	0.483*** (0.111)
Buffer × 1 [Bidder WACC > Target WACC]	-0.319*** (0.130)	-0.347*** (0.122)	-0.377*** (0.130)	-0.418*** (0.126)	-0.562*** (0.156)
1 [Bidder WACC > Target WACC]	0.012 (0.009)	0.015* (0.009)	0.017* (0.010)	0.021** (0.009)	0.024** (0.010)
Target WACC				0.086 (0.096)	0.190* (0.117)
Bidder WACC				-0.244** (0.114)	-0.311** (0.140)
Constant	-0.025*** (0.004)	-0.028*** (0.004)	-0.031*** (0.005)	-0.029*** (0.005)	-0.033*** (0.005)
Observations	746	746	736	736	625
R-squared	0.009	0.071	0.190	0.223	0.281
Year FE	Yes	Yes	Yes	Yes	Yes
Bidder & Target Industry FE					
Bidder × Target Industry FE					
Bidder & Target Controls					

- Table 2, Panel A shows that higher implied bidder buffers are associated with lower acquisition premia and higher bidder announcement returns.
- The premium coefficients are about -1.46 and -1.22 , while the bidder-CAR coefficients are about 0.21 to 0.22 .
- A 5 percentage point larger buffer predicts about a 1.1 percentage point higher bidder CAR in the richest baseline specification.
- Panel B is just as important: the target CAR is not significantly related to the buffer, and combined-firm CAR is also not significantly related to the buffer.
- But bidder share of dollar abnormal returns rises significantly with the buffer (coefficients around 1.37 and 1.23).
- This pattern strongly supports the bargaining interpretation: the buffer shifts the split of gains, rather than identifying better deals.
- Table 3 further shows that this effect is strongest when the bidder has a *smaller* WACC than the target and thus weaker ex ante bargaining power in the model's terms. The interaction coefficient is negative, so the marginal effect of the buffer is largest exactly where the model predicts.

5.6 Tables 4–6 and Figure 6: ex ante bargaining power and buffer use

Read:

- Table 4 uses supplier concentration as a measure of weak bargaining power. A one-standard-deviation increase in supplier concentration raises the probability of using a buffer by about 3–4 percentage points and raises the intensive margin by about 40–50 basis points.
- Table 5 uses customer concentration. The extensive-margin effect is weak, but the intensive-margin effect is positive and economically meaningful: about 0.33 to 0.46 percentage points for a one-standard-deviation increase.
- Table 6 uses supplier relative markup. Again the signs fit the model. A one-standard-deviation increase is associated with roughly a 3 percentage point increase in the probability of using a

Table 4

The buffer and supplier concentration.

This table explores how the bargaining power of a firm over its suppliers affects the buffer. If a company's suppliers come from concentrated industries, all else equal the company has less bargaining power. Our measure of supplier bargaining power is supplier concentration: the average of a firm's suppliers' industry-level sales concentration (Auer et al., 2020). We use Revere data on customer-supplier relationships to identify a firm's major suppliers. We aggregate to NACCS-3 by year level and match to our CFO survey data by their industries. Columns 1-3 focus on the extensive margin; columns 4-6 focus on the intensive margin. The variables Supplier Concentration, Beta Volatility and Sales Volatility are standardized to mean zero, unit variance. All variables are defined in detail in Table B.1. Table IA.4 provides robustness where we use different measures of industry-level concentration. Standard errors are clustered at survey industry $\times$ survey quarter and displayed in parentheses below the coefficient. ***, **, * denote significance at 1%, 5%, 10%.

	(1)	(2)	(3)	(4)	(5)	(6)
	Extensive margin			Intensive margin		
Supplier concentration	0.038*** (0.014)	0.034** (0.014)	0.030*** (0.013)	0.473*** (0.143)	0.426*** (0.132)	0.505*** (0.139)
Cost of capital	-0.022*** (0.003)	-0.022*** (0.003)	-0.028*** (0.003)	-0.312*** (0.046)	-0.320*** (0.046)	-0.313*** (0.047)
Beta volatility	0.027*** (0.012)	0.027*** (0.012)	0.027*** (0.012)	0.478*** (0.168)	0.454*** (0.169)	0.454*** (0.169)
Sales volatility	0.012 (0.016)	0.012 (0.016)	0.012 (0.016)	0.451*** (0.149)	0.396** (0.156)	0.396** (0.156)
Observations	1066	1066	1066	1066	1066	1066
R-squared	0.086	0.091	0.121	0.070	0.084	0.112
Survey quarter FE			Yes			Yes
Size FE			Yes			Yes
Ownership FE			Yes			Yes
Credit rating FE			Yes			Yes

Table 6

The buffer and the relative markup of suppliers.

This table explores how the relative bargaining power of a firm's suppliers affects the buffer. To measure the relative bargaining power of suppliers, we use the average ratio of supplier firms' markups to the firm's own markup over its customers. We use Revere data on customer-supplier relationships to identify a firm's major suppliers, and we use the "accounting" markup from *Baigues and Parls (2020)* as a proxy for the firm's bargaining power. For a given firm i , the Supplier Relative Markup is the simple average of the ratio of each supplier's markup to the firm's own markup over its customer. We aggregate to NACCS-3 by year level and match to our CFO survey data by their industries. Columns 1-3 focus on the extensive margin; columns 4-6 focus on the intensive margin. The variables Supplier Relative Markup, Beta Volatility and Sales Volatility are standardized to mean zero, unit variance. All variables are defined in detail in Table B.1. Standard errors are clustered at survey industry $\times$ survey quarter and displayed in parentheses below the coefficient. ***, **, * denote significance at 1%, 5%, 10%.

	(1)	(2)	(3)	(4)	(5)	(6)
	Extensive margin			Intensive margin		
Supplier relative markup	0.054*** (0.017)	0.028* (0.016)	0.028* (0.016)	0.532*** (0.157)	0.388** (0.157)	0.367** (0.163)
Cost of capital	-0.028*** (0.003)	-0.028*** (0.003)	-0.028*** (0.003)	-0.324*** (0.046)	-0.327*** (0.047)	-0.322*** (0.047)
Beta volatility	0.027*** (0.012)	0.027*** (0.012)	0.027*** (0.012)	0.475*** (0.167)	0.467*** (0.167)	0.467*** (0.167)
Sales volatility	0.005 (0.015)	0.005 (0.015)	0.007 (0.015)	0.365** (0.148)	0.322** (0.158)	0.322** (0.158)
Observations	1066	1066	1066	1066	1066	1066
R-squared	0.084	0.088	0.121	0.072	0.083	0.108
Survey quarter FE			Yes			Yes
Size FE			Yes			Yes
Ownership FE			Yes			Yes
Credit rating FE			Yes			Yes

Table 5

The buffer and customer concentration.

This table explores how the bargaining power of a firm's customers affects the buffer. If a company's customers are concentrated, all else equal the company has less bargaining power; and vice versa. Our measure of customer bargaining power is customer concentration: the HHI of a firm's sales to corporate customers (Pattabai, 2012). We aggregate to NACCS-3 by year level and match to our CFO survey data by their industries. Columns 1-3 focus on the extensive margin; columns 4-6 focus on the intensive margin. The variables Customer Concentration, Beta Volatility and Sales Volatility are standardized to mean zero, unit variance. All variables are defined in detail in Table B.1. Standard errors are clustered at survey industry $\times$ survey quarter and displayed in parentheses below the coefficient. ***, **, * denote significance at 1%, 5%, 10%.

	(1)	(2)	(3)	(4)	(5)	(6)
	Extensive margin			Intensive margin		
Customer concentration	0.009 (0.015)	0.004 (0.014)	0.006 (0.015)	0.461*** (0.148)	0.333*** (0.148)	0.352*** (0.148)
Cost of capital	-0.022*** (0.003)	-0.022*** (0.003)	-0.028*** (0.003)	-0.316*** (0.047)	-0.322*** (0.047)	-0.317*** (0.047)
Sales volatility	0.009 (0.016)	0.009 (0.016)	0.019 (0.016)	0.366** (0.146)	0.366** (0.146)	0.313** (0.154)
Beta volatility	0.022*** (0.012)	0.022*** (0.012)	0.022*** (0.012)	0.452*** (0.164)	0.452*** (0.164)	0.450*** (0.164)
Observations	1066	1066	1066	1066	1066	1066
R-squared	0.078	0.084	0.117	0.069	0.081	0.107
Survey quarter FE			Yes			Yes
Size FE			Yes			Yes
Ownership FE			Yes			Yes
Credit rating FE			Yes			Yes

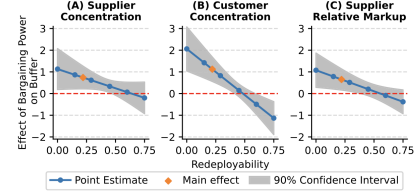


Fig. 6. The relation between the buffer and bargaining power by asset specificity.

This figure displays how the relations between the buffer and our measures of bargaining power vary across the degree of industry-level asset redeployability. Firms are more likely to bargain over assets that more specific-to-use, or less redeployable, so the relation between the buffer and bargaining power should be stronger in low-redeployability industries. For bargaining power measures, we estimate a regression of the buffer (intensive margin) on an interaction between bargaining power and industry level redeployability, using the same fixed effects and controls as in Tables 4-6. We then plot the predicted effect of a one-standard deviation increase in bargaining power BP on the buffer, conditional on redeployability. In each plot, we also display the main effect from the regression, when redeployability is at the mean. Standard errors are clustered by survey industry $\times$ survey quarter. Table IA.7 Panel A displays the regressions used to estimate each figure (columns 4-6 map to this figure).

buffer and a 0.37 to 0.53 percentage point increase in buffer size.

- **Figure 6** adds a powerful interaction: the relation between bargaining power and the buffer is stronger when assets are less redeployable and therefore more specific to the relationship.
- The paper highlights one concrete magnitude: the effect of a one-standard-deviation increase in supplier concentration on the buffer is about 0.9 percentage points larger at 25% redeployability than at 75% redeployability.

6 Short summary

- **Method.** The paper combines a delegated-bargaining theory of hurdle-rate choice with CFO survey data, ROIC bunching evidence, M&A fairness-opinion discount rates, and customer-supplier bargaining-power proxies.
- **Identification.** The bargaining channel is identified from patterns consistent with surplus capture rather than deal selection: lower premia and higher bidder returns without higher combined gains, stronger effects when ex ante bargaining power is weaker, and stronger buffer-bargaining relations when asset specificity is high.
- **Application.** Firms use large and persistent hurdle-rate buffers. These buffers are organizationally meaningful: they are sticky, set from the top, and realized returns bunch just above them.
- **Main result.** Elevated hurdle rates can preserve or enhance firm value by improving bargaining outcomes, even though they also cause firms to reject some positive-NPV projects.

7 Conclusion

7.1 Core conclusion

- The paper shows that hurdle-rate buffers have an economically important consequence beyond simple project screening: they affect negotiated surplus.
- It therefore challenges the standard claim that buffers are necessarily evidence of underinvestment and value destruction.
- In both theory and data, commitment to a higher hurdle can be privately valuable because it strengthens the firm's walkaway position.

7.2 Economic intuition

- A hurdle rate is not just a discounting convention. Inside the firm, it is an instruction passed down from the CFO to delegates negotiating with outsiders.
- Once that instruction is stable and credible, it becomes a bargaining tool. Counterparties understand that the delegate cannot easily accept terms that do not clear the hurdle.
- This improves the firm's terms but also narrows the set of acceptable projects. The optimal buffer balances those two effects.
- Firms with weak ex ante bargaining power or with more relationship-specific assets benefit more from this commitment technology.

7.3 Takeaway

This paper is important because it connects three literatures that are usually kept separate: capital budgeting, bargaining, and corporate asset acquisition. Its main message is not that traditional reasons for hurdle-rate buffers are wrong, but that their consequences have been misunderstood. A high hurdle rate can help firms obtain better deals, which means that the persistence of large buffers may be rational even when the textbook NPV logic alone would suggest otherwise.